Troubleshooting

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Troubleshooting

Troubleshooting

You might encounter a failure while upgrading the node pair. The node might crash, aggregates might not relocate, or LIFs might not migrate. The cause of the failure and its solution depend on when the failure occurred during the upgrade procedure.

Refer to the table describing the different phases of the procedure in the section Overview of the ARL upgrade. Information about the failures that can occur is listed by the phase of the procedure.

Aggregate relocation failures

Aggregate relocation (ARL) might fail at different points during the upgrade.

Checking for aggregate relocation failure

During the procedure, ARL might fail in Stage 2, Stage 3, or Stage 5.

Steps

1. Enter the following command and examine its output:

   `storage aggregate relocation show`

   The `storage aggregate relocation show` command shows you which aggregates were successfully relocated and which ones were not, along with the causes of failure.

2. Check the console for any EMS messages.
3. Take one of the following actions:
   - Take the appropriate corrective action, depending on the output of the `storage aggregate relocation show` command and the output of the EMS message.
   - Force relocation of the aggregate or aggregates by using the `override-vetoes` option or the `override-destination-checks` option of the `storage aggregate relocation start` command.

     For detailed information about the `storage aggregate relocation start`, `override-vetoes`, and `override-destination-checks` options, see the ONTAP 9.8 Commands: Manual Page Reference.

Aggregates originally on node1 are owned by node4 after completion of the upgrade

At the end of the upgrade procedure, node3 should be the new home node of aggregates that originally had node1 as the home node. You can relocate them after the upgrade.

About this task

Aggregates might fail to relocate properly, that is, they have node1 as their home node instead of node3, under the following circumstances:

- During Stage 3, when aggregates are relocated from node2 to node3.

  Some of the aggregates being relocated have node1 as their home node. For example, such an aggregate could be called aggr_node_1. If relocation of aggr_node_1 fails during Stage 3, and relocation cannot be forced, then the aggregate is left behind on node2.

- After Stage 4, when node2 is replaced with node4.

  When node2 is replaced, aggr_node_1 will come online with node4 as its home node instead of node3.

  You can fix the incorrect ownership problem after Stage 6, after you have enabled storage failover by completing the following steps:

Steps

1. Get a list of aggregates by using the following command:

   ```
   storage aggregate show -nodes <node4> -is-home true
   ```

   To identify aggregates that were not correctly relocated, refer to the list of aggregates with the home owner of node1 that you obtained in the section Prepare the nodes for upgrade and compare it with the output of the above command.

2. Compare the output of Step 1 with the output you captured for node1 in the section Prepare the nodes for upgrade and note any aggregates that were not correctly relocated.
3. Relocate the aggregates left behind on node4 by using the following command:

   ```
   storage aggregate relocation start -node <node4> - aggr <aggr_node_1> -destination <node3>
   ```

   Do not use the `-ndo-controller-upgrade` parameter during this relocation.

4. Verify that node3 is now the home owner of the aggregates by using the following command:

   ```
   storage aggregate show -aggregate <aggr1,aggr2,aggr3…> -fields home-name
   ```

   `<aggr1,aggr2,aggr3…>` is the list of aggregates that had node1 as the original home owner.

   Aggregates that do not have node3 as home owner can be relocated to node3 using the same relocation command in Step 3.

Reboots, panics, or power cycles

The system might crash – reboot, panic or go through a power cycle – during different stages of the upgrade.

The solution to these problems depends on when they occur.

Reboots, panics, or power cycles during the pre-check phase

Node1 or node2 crashes before the pre-check phase with HA pair still enabled

If either node1 or node2 crashes before the pre-check phase, no aggregates have been relocated yet and the HA pair configuration is still enabled.

About this task

Takeover and giveback can proceed normally.

Steps

1. Check the console for EMS messages that the system might have issued and take the recommended corrective action.
2. Continue with the node-pair upgrade procedure.

Reboots, panics, or power cycles during first resource-release phase

Node1 crashes during the first resource-release phase with HA pair still enabled

Some or all aggregates have been relocated from node1 to node2, and HA pair is still enabled. Node2 takes over node1's root volume and any non-root aggregates that were not relocated.

About this task

Ownership of aggregates that were relocated look the same as the ownership of non-root aggregates that were taken over because the home owner has not changed.

When node1 enters the `waiting for giveback` state, node2 gives back all of the node1 non- root aggregates.

Steps

1. After node1 is booted up, all the non-root aggregates of node1 have moved back to node1. You must perform a manual aggregate relocation of the aggregates from node1 to node2 by using the following command:
   `storage aggregate relocation start -node node1 -destination node2 -aggregate -list * - ndocontroller-upgrade true`
2. Continue with the node-pair upgrade procedure.

Node1 crashes during the first resource-release phase while HA pair is disabled

Node2 does not take over but it is still serving data from all non-root aggregates.

Steps

1. Bring up node1.
2. Continue with the node-pair upgrade procedure.

Node2 fails during the first resource-release phase with HA pair still enabled

Node1 has relocated some or all of its aggregates to node2. The HA pair is enabled.

About this task

Node1 takes over all of node2's aggregates as well as any of its own aggregates that it had relocated to node2.

When node2 boots up, the aggregate relocation is completed automatically.

Steps

1. Bring up node2.
2. Continue with the node-pair upgrade procedure.

Node2 crashes during the first resource-release phase and after HA pair is disabled

Node1 does not take over.

Steps

1. Bring up node2.

   A client outage occurs for all aggregates while node2 is booting up.

2. Continue with rest of the node-pair upgrade procedure.

Reboots, panics, or power cycles during the first verification phase

Node2 crashes during the first verification phase with HA pair disabled

Node3 does not take over following a node2 crash as the HA pair is already disabled.

Steps

1. Bring up node2.

   A client outage occurs for all aggregates while node2 is booting up.

2. Continue with the node-pair upgrade procedure.

Node3 crashes during the first verification phase with HA pair disabled

Node2 does not take over but it is still serving data from all non-root aggregates.

Steps

1. Bring up node3.
2. Continue with the node-pair upgrade procedure.

Reboots, panics, or power cycles during first resource-regain phase

Node2 crashes during the first resource-regain phase during aggregate relocation

Node2 has relocated some or all of its aggregates from node1 to node3. Node3 serves data from aggregates that were relocated. The HA pair is disabled and hence there is no takeover.

About this task

There is client outage for aggregates that were not relocated. On booting up node2, the aggregates of node1 are relocated to node3.

Steps

1. Bring up node2.

2. Continue with the node-pair upgrade procedure.

Node3 crashes during the first resource-regain phase during aggregate relocation

If node3 crashes while node2 is relocating aggregates to node3, the task continues after node3 boots up.

About this task

Node2 continues to serve remaining aggregates, but aggregates that were already relocated to node3 encounter client outage while node3 is booting up.

Steps

1. Bring up node3.
2. Continue with the controller upgrade.

Reboots, panics, or power cycles during post-check phase

Node2 or node3 crashes during the post-check phase

The HA pair is disabled hence this is no takeover. There is a client outage for aggregates belonging to the node that rebooted.

Steps

1. Bring up the node.
2. Continue with the node-pair upgrade procedure.

Reboots, panics, or power cycles during second resource-release phase

Node3 crashes during the second resource-release phase

If node3 crashes while node2 is relocating aggregates, the task continues after node3 boots up.

About this task

Node2 continues to serve remaining aggregates but aggregates that were already relocated to node3 and node3's own aggregates encounter client outages while node3 is booting.

Steps

1. Bring up node3.
2. Continue with the controller upgrade procedure.

Node2 crashes during the second resource-release phase

If node2 crashes during aggregate relocation, node2 is not taken over.

About this task

Node3 continues to serve the aggregates that have been relocated, but the aggregates owned by node2 encounter client outages.

Steps

1. Bring up node2.
2. Continue with the controller upgrade procedure.

Reboots, panics, or power cycles during the second verification phase

Node3 crashes during the second verification phase

If node3 crashes during this phase, takeover does not happen because the HA pair is already disabled.

About this task

There is a client outage for all aggregates until node3 reboots.

Steps

1. Bring up node3.
2. Continue with the node-pair upgrade procedure.

Node4 crashes during the second verification phase

If node4 crashes during this phase, takeover does not happen. Node3 serves data from the aggregates.

About this task

There is an outage for non-root aggregates that were already relocated until node4 reboots.

Steps

1. Bring up node4.
2. Continue with the node-pair upgrade procedure.

Issues that can arise in multiple stages of the procedure

Some issues can occur during different stages of the procedure.

Unexpected "storage failover show" command output

During the procedure, if the node that hosts all data aggregates panics or is rebooted accidentally, you might see unexpected output for the `storage failover show` command before and after the reboot, panic, or power cycle.

About this task

You might see unexpected output from the `storage failover show` command in Stage 2, Stage 3, Stage 4, or Stage 5.

The following example shows the expected output of the `storage failover show` command if there are no reboots or panics on the node that hosts all the data aggregates:

```
cluster::> storage failover show

                    Takeover
Node      Partner   Possible  State Description
-------  --------  --------- -----------------
node1     node2     false     Unknown
node2     node1     false     Node owns partner aggregates as part of the
non-disruptive head upgrade procedure. Takeover is not possible: Storage
failover is disabled.
```

The following example shows the output of the `storage failover show` command after a reboot or panic:

```
cluster::> storage failover show

                    Takeover
Node      Partner   Possible  State Description
-------  --------  --------- -----------------
node1     node2     -         Unknown
node2     node1    false      Waiting for node1, Partial giveback, Takeover
is not possible: Storage failover is disabled
```

Although the output says that a node is in partial giveback and that storage failover is disabled, you can disregard this message.

Steps

No action is required; continue with the node-pair upgrade procedure.

LIF migration failure

After you migrate LIFs, they might not come online after migration in Stage 2, Stage 3, or Stage 5.

Steps

1. Verify that the port MTU size is the same as that of the source node.

   For example, if the cluster port MTU size is 9000 on the source node, it should be 9000 on the destination node.

2. Check the physical connectivity of the network cable if the physical state of the port is `down`.

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